

L2 - 12.6. The Silent Transfer

Case Briefing

This challenge assumes you can filter packet captures, work with Zeek logs, and pivot between related network events.

The Investigation

A forensic workstation is available for this investigation. It includes Wireshark, TShark, Zui, Zeek command-line tools such as `zeek-cut` (extracts specified columns from Zeek's tab-separated log files.), and standard terminal utilities. All evidence is stored in `/home/ubuntu/capstone/`:

- `snort_alerts.log` : Snort detection output
- `zeek_logs/` : Zeek connection, DNS, TLS, HTTP, file, and notice logs
- `investigation.pcap` : Packet capture for packet-level validation
- `fortigate_traffic.log` : Firewall traffic covering internal and cross-subnet activity
- `references/` : Local threat intelligence and MITRE ATT&CK reference material

SSL (Secure Sockets Layer) and SMB (Server Message Block)

Log File	Description	Key Information Captured
<code>conn.log</code>	Connection logs	Overall summary of all TCP/UDP/ICMP network connections (source/destination IPs, ports, duration, bytes sent/received).
<code>dns.log</code>	DNS activity	All Domain Name System queries and responses (domain names requested, resolved IP addresses, query types).
<code>files.log</code>	File transfers	Metadata for files transferred over network protocols like HTTP, FTP, or SMB (hashes, file size, MIME type).
<code>http.log</code>	HTTP traffic	Details of plain-text web requests and responses (URLs visited, user-agents, HTTP methods, status codes).

<code>notice.log</code>	Security alerts	Potential anomalies, policy violations, or suspicious behaviors flagged by Zeek's detection scripts.
<code>ssl.log</code>	TLS/SSL handshakes	Encrypted connection metadata (SSL versions, cipher suites, server certificate details, SNI/domain names).

Answer the questions below

Review the detection evidence around 03:47 UTC and correlate it with the repeated C2 traffic. Which internal IP address originated that traffic?

10.14.30.88

```
ubuntu@tryhackme:~$ cd /home/ubuntu/capstone/
ubuntu@tryhackme:~/capstone$ ls
README.txt          investigation.pcap  snort_alerts.log
fortigate_traffic.log references          zeek_logs
ubuntu@tryhackme:~/capstone$ cat README.txt
Operation SILENT TRANSFER – Capstone Investigation
=====

Storyline: A Snort alert has fired on WKST-DEV-12 (10.14.30.88) in the
Developer subnet at 03:47:22 UTC on 2025-11-14. The alert message is
"ET TROJAN Possible Cobalt Strike Beacon CnC Activity - GET Checkin".
```

Working backwards from the C2 activity, which domain was used to deliver the initial dropper to the compromised workstation?

<http://cdn-updates.microsoftservice.net/>

```
ubuntu@tryhackme:~/capstone/zeek_logs$ cat dns.log
```

Identify the file downloaded from the delivery domain. What is its SHA256 hash?

7f3b2e1a9c8d4f5e6b7c8d9e0f1a2b3c4d5e6f7a8b9c0d1e2f3a4b5c6d7e8f90

```
ubuntu@tryhackme:~/capstone/zeek_logs$ cat files.log
```

Which source port did the compromised workstation use for its first connection to the C2 server?

51000

Open http.logs with nano

```
File Edit View Search Terminal Help
GNU nano 7.2 http.log
#separator (add)
#set_separator (empty)
#empty_field (empty)
#unset_field
#path http
#open 2025-11-14 00:00:00
#fields ts uid id.orig_h id.orig_p id.resp_h id.resp_p trans_depth method host url referrer version user_agent
#types time string addr port addr port count string string string string string count count count string count str
1763081220.000000 CrD5UMpFW617XYhqtK 10.14.30.88 52100 194.165.16.78 80 1 GET 194.165.16.78 /update/winservice-patch-4891.exe
1763083380.000000 CDH4vog1ELRT522A0A 10.14.30.88 51000 194.165.16.56 443 1 GET update.softpatch-cdn.com /api/v2/check -
1763083442.385990 CM08L9p25yB10wdXIs 10.14.30.88 51001 194.165.16.56 443 1 GET update.softpatch-cdn.com /api/v2/check -
1763083498.766728 CQt1xh9Dt33W4RZcn3 10.14.30.88 51002 194.165.16.56 443 1 GET update.softpatch-cdn.com /api/v2/check -
1763083561.694029 CcNr0isp2ropFbTg7s 10.14.30.88 51003 194.165.16.56 443 1 GET update.softpatch-cdn.com /api/v2/check -
1763083621.065753 CsykaT3oJex90oa0ca 10.14.30.88 51004 194.165.16.56 443 1 GET update.softpatch-cdn.com /api/v2/check -
```

Review the TLS activity between the compromised workstation and the C2 server. What JA4 fingerprint identifies the C2 client?

t13d190900_9dc949149365_97f8aa674fd9

```
ubuntu@tryhackme:~/capstone/zeek_logs$ ls
conn.log  dns.log  files.log  http.log  notice.log  ssl.log
```

Open ssl.logs with nano , look for self signed certificate

After C2 was established, how many unique internal destination IP addresses did the compromised workstation contact during its SMB discovery activity?

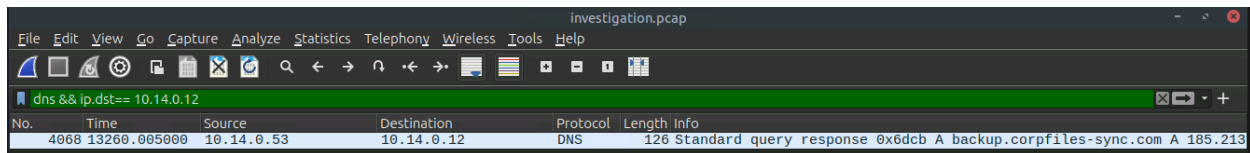
23

```
ubuntu@tryhackme:~/capstone/zeek_logs$ zeek-cut id.orig_h id.
resp_h id.resp_p < conn.log | awk -F'\t' '$1=="10.14.30.88"
&& $3=="445"' | cut -f2 | sort -u | wc -l23
```

Following the SMB activity, the attacker established an RDP connection to an internal server. What is the destination IP address?

10.14.0.12

```
dns && ip.dst== 10.14.0.12
```



The image shows a Wireshark packet capture window titled 'Investigation.pcap'. The filter bar contains 'dns && ip.dst== 10.14.0.12'. The packet list shows a single packet at time 4068.13260.005000, source 10.14.0.53, destination 10.14.0.12, protocol DNS, length 126. The packet details show a 'Standard query response' from 0x6dcb A backup.corpfiles-sync.com A 185.213.

No.	Time	Source	Destination	Protocol	Length	Info
4068	13260.005000	10.14.0.53	10.14.0.12	DNS	126	Standard query response 0x6dcb A backup.corpfiles-sync.com A 185.213

Review the DNS activity originating from the RDP destination. Which domain did the server resolve immediately before the large outbound transfer?

backup.corpfiles-sync.com

```
ubuntu@tryhackme:~/capstone/zeek_logs$ zeek-cut ts id.orig_h  
query answers < dns.log \  
| awk -F'\t' '$2=="10.14.0.12" && $1<1763091720' | sort -k1 -  
n | tail -5
```

```
ubuntu@tryhackme:~/capstone/zeek_logs$ zeek-cut ts id.orig_h query answers < dns.log \  
| awk -F'\t' '$2=="10.14.0.12" && $1<1763091720' | sort -k1 -n | tail -5  
1763089918.000000      10.14.0.12      update.ubuntu.com      91.189.91.83  
1763090193.000000      10.14.0.12      docker.io              104.21.14.57  
1763090355.000000      10.14.0.12      cdn.jsdelivr.net       104.16.132.229  
1763090454.000000      10.14.0.12      update.ubuntu.com       91.189.91.83  
1763091660.000000      10.14.0.12      backup.corpfiles-sync.com 185.213.154.201
```

Identify the archive transferred from the internal server to the external endpoint. What is its SHA256 hash?

a3f8e2c1d4b7a9e0f2c3d5e6f7a8b9c0d1e2f3a4b5c6d7e8f9a0b1c2d3e4f5a6

```
ubuntu@tryhackme:~/capstone/zeek_logs$ cat files.log  
#separator \x09  
#set_separator ,  
#empty_field (empty)  
#unset_field -  
#path files  
#open 2025-11-14-06-00-00  
#fields ts      fuid      tx_hosts      rx_hosts      conn_uids      source      depth      analyzers      mime_type      filename      duration      local_orig  
ts_orig seen_bytes total_bytesmissing_bytes overfow_bytes timeout parent_fuid md5 sha1 sha256 extracted extracted_cutoff extracted_sl  
ze  
#types time      string      set[addr]      set[addr]      set[string]      string      count      set[string]string      string      interval      bool      bool      count      count  
count count bool string string string string string bool count  
1763081220.000000 FHGphAntbrPGRfMmu9 194.165.16.78 10.14.30.88 CrD5UMpFW617XyhgTK HTTP 0 SHA256,MD5,SHA1 application/x-dosexec winservi  
ce-patch-4891.exe 8.500000 FF 1887232 1887232 0 0 F a1b2c3d4e5f6a7b8c9d0e1f2a3b4c5d6 abcdef1234567890abcdef1234567890  
abcdef12 7f3b2e1a9c8d4f5e6b7c8d9e0f1a2b3c4d5e6f7a8b9c0d1e2f3a4b5c6d7e8f90 - F 0  
1763091720.000000 FdD9J8z0g5eQH7NYx 10.14.0.12 185.213.154.201 CHQztwtwi2Jkq87YgB HTTP 0 SHA256,MD5,SHA1 application/zip Q4-Finance-Backu  
p-2025.zip 480.000000 T T312447821 312447821 0 0 F b2c3d4e5f6a7b8c9d0e1f2a3b4c5d6e7 1234567890abcdef1234567890abc  
def12345678 b3f8e2c1d4b7a9e0f2c3d5e6f7a8b9c0d1e2f3a4b5c6d7e8f9a0b1c2d3e4f5a6 - F 0  
#close 2025-11-14-06-00-00
```

Inspect the application-layer contents of the C2 traffic. What command did the attacker issue to the compromised workstation?

whoami

```
ubuntu@tryhackme:~/capstone$ strings investigation.pcap | grep "cmd"
{"status":"ok","cmd":"","interval":60}t
{"status":"ok","cmd":"","interval":60}
{"status":"ok","cmd":"","interval":60}
{"status":"ok","cmd":"","interval":60}"
{"status":"ok","cmd":"","interval":60}\
{"status":"ok","cmd":"d2hvYW1p","interval":60}
{"status":"ok","cmd":"","interval":60}
```

decode d2hvYW1p

The screenshot shows a web-based Base64 decoder interface. On the left, the 'Recipe' panel is configured with 'From Base64' selected, an 'Alphabet' dropdown set to 'A-Za-z0-9+/', and the 'Remove non-alphabet chars' checkbox checked. The 'Input' field on the right contains the Base64 string 'd2hvYW1p'. Below the input, the 'Output' field displays the decoded text 'whoami'. The interface includes various icons for saving, copying, and viewing raw bytes.